

Active Inference Stream #010.1: "A variational approach to scripts" (2020)

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hello everyone and welcome to the active inference live stream this is active inference live stream 10.1 and it is december 8th 2020 we are really excited to bring you a scripted or unscripted not sure maybe we'll figure it out discussion on some exciting topics so let's get right into it i will share my screen for the participants welcome to the active inference lab everyone we are an experienced experiment and experience i guess in online team communication learning and practice related to active inference you can find us at our website on twitter at our email address our youtube channel or public key based team and username this is a recorded and an archived live stream so please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome here and just as far as video etiquette for live stream goes remember to mute if there's noise in your background raise your hand so that we can hear from everyone on the stack and use respectful speech behavior also just a logistical note all the rest of the meetings for 2020 are going to be at 7 30 to 9 a.m pst and today we are in 10.1 discussing a variational approach to scripts next week at the same time we're going to be having a follow-up discussion 10.2 and then in the last two weeks of december we're going to be talking about the paper sophisticated effective inference and that will be on the december 22nd and 29th so contact us if you want to participate in these discussions or in 2021 or if you have an idea for something that we could do on an active stream today in act in stream 10.1 we are going to go through introductions and warmups and then we'll go to template one we're going to be discussing the paper a variational approach to scripts and if you want some background context you can check out active stream 10.0 we're going to talk today about the aims and claims or the hopes as they were and the abstract and the roadmap for the paper thankfully with one of the authors on board and then we'll talk about some different kinds of scripts talk about scripts and their uses ask a few questions related to that and then close with looking through the figures and just discussing what they might mean and next week in 10.2 we're going to be on this same paper so save and submit your questions

or if it's after 10.2 just write them in the comment section and get in touch if you want to participate for the introductions and warmups let's introduce ourselves just by giving a short introduction or check-in and then we'll pass it to somebody who hasn't spoken yet so i'll start i'm daniel i'm excited for this conversation and i will pass it to our first time visitor mao hi so yes my name is mao alba i'm one of the authors on the paper along with axel constance maxwell ramstead and karl fristen i'm a first year phd student in quebec and i'm working in cognitive computing uh to try to establish how we can formalize epistemic communities through active cool then we'll pass to another first time guest rj sure uh so my name is rj corday uh i'm in new york and i'm currently researching organizational dynamics intelligence analysis and narrative warfare keep it short about sasha thanks hi i'm sasha i'm based out of california and i'm also a graduate student studying developmental neuroscience with that i'll pass it on to blue blue muted we can hear you awesome thank you i'm blue knight i'm from new mexico and i am an independent research consultant and i'm currently working on agent-based modeling and a bunch of other random things i'll pass it to shannon hey i'm shannon i'm typically based out of california i'm also a phd student i'm interested in music neuroscience or i work on music neuroscience and i'm interested in music and social interactions so i was really excited for that social side of this paper and i'll pass it to yvonne hello my name is ivan i'm based in moscow and i'm interested in active inference as it goes with the systems engineering and i'm very excited about this paper i found it uh very interesting cool thanks everyone for coming out and for everybody who's watching it live and replay so for the warm-up questions section everybody is welcome to throw in a thought i'll put up the uh first two questions and just raise your hand so that we can hear from anyone the first question is what interests you or excites you about studying scripts so kind of what drew you to this paper whether from the active inference side or from the social sciences side and then what's something that you thought about after reading the paper so while people are raising their hand i'll start i think what interested me in uh studying scripts was just addressing all of these different social contexts with one integrated framework that's kind of an appealing idea mao well yeah and i think it's also it's sort of a road map that links internal and external manifestations of our general understanding of the world and systems dynamics that get um brought together through these islands of negentropy that become these scripts if that makes any sense cool rj um so uh after i read the paper what i ended up thinking about is what do the conditions look like for threshold events for changing scripts um that's a really interesting to me and then um what kind of like what what would the metadata be and this is

something we had discussed uh before is like what would the metadata be if we wanted to catalog the scripts that are that are operating in a system cool and that um bringing up that negentropic idea the sort of organizing principle of scripts that's pretty interesting too because uh in combining what both of you just said how do we change a script from one island of stability or organizational attractor to another organizational attractor if we go from one organizing principle maybe we don't like some aspects of it but if we just go from order to pure disorder that's not doing the society of service so we want to move from order to order ideally through order but how is that actually going to be uh tracked or designed and then just the last up shannon go ahead yeah that's actually exactly the one question i had written on my paper um are is what age rj and um daniel mentioned like what can active inference help us to identify like when a critical transition will happen or when we'll switch scripts um it's really easy in it's really easy i say but in in music we have scripts and it's either written on a piece of paper or it's the cultural transmission of how you perform in this genre of music alone or with others and as long as you know that that's the script that you're starting from you know how the social interaction will transpire and then if you want to do an experiment you're like you know that's the constraints that you have and you can pick out the interesting features that you want to empirically measure like whether it's body sway or the actual music that's being played but um in the example that runs through the paper of flirting you don't necessarily know the goal of the interaction if you're observing it or the goal of the interaction if you're in it because it's not codified in like this very formal or traditional framework in the same way that like a musical interaction is so identifying the threshold or identifying how those transitions happen would be very different in those two types of interactions cool blue so i was excited to really learn more about um how scripts can interact in a hierarchical way like just the paper kind of mentioned it but didn't really unpack it uh thoroughly and i just would like to really kind of dive into that yeah and the music and the cultural side it's uh maybe with a musical ensemble they could agree to play a different song within the same style or agree to play a different style and then we could ask the same with culture but it's a little bit more challenging to simply adopt a different cultural regime but maybe within a culture there's a lot of latitude so there's this question of nesting and horizontal interactions and about style controlled novelty so i think there's a lot of cool ideas we'll be bringing up any other comments here on these warm-up questions mal up you're me too sorry so yeah that's the idea that's the goal the um eventually measuring where the threshold happens um we have another paper that's being written and that's trying to identify how scripts tend to become overlapping

through in interaction in society and when they become overlapping they they become analogical so um the scripts of femininity and masculinity are not as segregated as they used to be for instance i'm wearing no nose sleeves in these 1800s but would definitely have been acceptable for men nowadays it's not a problem anymore both men and women can wear it so these these scripts have tended to overlap and we're seeing that because the scripts overlap they become a little bit meaningless because they're not allowing us to predict anything so there's going to be to be a point which we're kind of sensing already where the concept of gender as a strict binary is just breaking apart and we're seeing new uh new concepts emerge new new possibilities so this this is where the threshold can be measured and i think we can definitely formalize that as far as codification um art in general is just a series of codes that once you have you get to break it's just it's become this sort of script in art where you have to break the script you you have to use what we know in order to continually create novelty but think of all of the hollywood movies that depict flirting it's pretty much the same like there's there's a few scripts that are possible but if you look at it abstractly it's basically you know you don't you don't flirt necessarily uh with someone you have zero intention of ever ever becoming with in hollywood movies or you don't flirt with someone by hitting them on the head with a flounder you know like like there's some things that you know you're supposed to do and some things you know you're not supposed to do and the strictness depends on how wide one vision of the script becomes thanks rj i just uh something i picked up from what you were saying there uh on um on flirting in in in cinema like i feel like a lot of the comedies i know of are actually about script breaking so even even like outside flirting like something like uh there's there's a like common category like in in in the uk it's peep show in the us it's always sunny um where it's most of the show is just about failing to adapt to cultural scripts so i just want to know like i feel like there's there's some element there where we actually like contrast like exactly what it is we're not supposed to do as well yep there's the exception that proves the rule and then there's the the chess master who within the structure of the rules makes an innovation that appears in the short term to be breaking the rules and then there's this element of the media which is showing us for example maybe flirting scenes with things that wouldn't work in real life uh whether for anyone or for subset of people but often including very for example bold maneuvers they can't uh convey in the hollywood reality meeting somebody getting to know them over years not as exciting not as a clear foreshadowing so maybe there are sort of rule or script breaking things but then when we see that representation and maybe even starting to introduce some of these active ideas we see them in our sensory

states and therefore we perceive them as likely or plausible and then we come to embody these beliefs about what is likely or possible or uh preferable in the world and we see people imitating behavior they see in the media so that can be used for good and for bad sasha yeah this kind of loops back to the question that i had after reading the paper which is um how do individuals convey um their true state or um their you know unique perspective on the script while conforming to the social script because um if we all acted like we were in a rom-com we would have different lives and relationships but clearly we all have very unique um and uh personalized um versions of that and i don't think any of us live in a rom-com but you know please correct me if i'm wrong yeah that's kind of the fun it's like asking what's the genre what scripts will my scripts uh or what genre will my scripts be drawn from am i in a tragedy am i in a war movie am i in a rom-com so that's kind of fun it's like what genre would your theme music be what kinds of conversants are you gonna be interacting with um mao did you have your hand raised and then aren't you yeah yeah go ahead so first thing is um i think you mentioned the the oppositeness and that's a really interesting idea like the that's also how new scripts emerge is that you have these these bubbles of counter discourse that come and that's either embody um embody portray something that is real in society but often invisibilized or something that um wants to push back on something that feels oppressive to a large swaths of society so for instance one great study that i saw i'm sorry about i i don't have the name of it right now but i'll send it to whoever wants it it was about the portrayal of abject in uk society and the point was there's all these mako versions that show you first all these women how they are not supposed to be so they take a woman and they portray her as very ugly often um lower class like they give her all the qualities you're not supposed to have and then they take her through this transition and they make over her life and suddenly at the end of the show she's happy her life is great her children love her her husband loves her again it's great and it's like this this movement from the abject that you're not supposed to embody to the to the ideal that you are supposed to embody and on the second note what you said i think sasha is that's exactly right but the thing is you are not the person in the rom-com you do not perfectly embody what she is even if they try to make her relatable you are connected through these tendrils of semantics to a variety of things that make you take what is possible and adapt all of these possible scripts into this version of adapted contextual anchored script which is why there is variety as well cool rj um i was gonna say uh mao covered a lot of what i was gonna say actually so we're we're fairly saying i was just gonna say like that um i like actually i like the reference to like makeover shows i was gonna i was gonna talk about um a very opposite

end uh but but uh greek art and stories where where it's it's very often and this is where like i think joseph campbell brings up the hero's journey um where it's from some state which is object negative not where we want to be to some other state none of us actually live in the rom-com like but but the the ideal being presented um isn't something we're supposed to mirror necessarily but something that kind of acts as a strange attractor um but yeah but anyway mal covered most of what i was gonna say there yeah rj i i wrote down literally hiro's journey because the makeover show if it's weight loss or about um norms for genders or house makeover or chefs it's always this hero's journey and it's a scripted hero's journey literally designed journey and then the interesting part is the policies that they undertake thinking about active those policies are not our affordances they have a team and they have a camera person and they have a coach so then it's like showing you the first point and the last point on the trajectory and saying well now you figure out what policy but the only example that you've observed the only example of what's a plausible affordance is something that is defined as being inaccessible to basically everyone so that's just like such a twisted way to play with how people actually improve themselves yet mao and then rj yeah so that's that's exactly right you you touched on a very important point is that take each script as though it was its own little like market blanket like it's not right but imagine um there's some parts of the scripts that will interact with one another and then there's these like blankets that move a little slower and through the blankets you can go to another part of the script and you can do this at all the levers so this was touching on the idea that someone else brought earlier that there's layers of scripts and that's absolutely true you can break down every script into smaller smaller scripts what defines a script is its edges so entering the and leaving the script what happens in between can also be scripted but that's at a lower level right so for me to for me to speak here i had to go through a series of things but what really matters is that i'm here now and that this will be recorded but um getting here it doesn't really matter how right unless i'm coming from it as a as an academic so i had a series of steps that i did to get there as an academic the the way that you bring these scripts together your vectors from where you're coming from define also what kinds of policies you'll have available to you inside a given script cool rj and then we'll continue uh yeah uh oh also um real quick i like that that's that's quotable what defines the script is its edges um but uh i just wanted to add real quick uh that it's probably actually dangerous to optimize too much to say like the rom-com or the ideal like we don't actually want to copy it exactly right like i feel like maybe maybe uh um there's there's something there where where we're supposed to actually mutate the scripts we're being presented over time and

that like if we actually optimize them exactly like a perfect mirror then we actually get instability but anyway cool well i hope if you're listening hearing about scripts for the first time or on the script side hearing about active inference you're starting to see where we are all getting excited about at this intersection the paper that we read today is a variational approach to scripts and the aims claims and hope of the paper were addressed in 10.0 so feel free to pause this video and recursively watch the other one if you want but mao since you're with us what would you say in your own words were the aims and claims or the intentions of the paper so the goal was to bring together fields that had different strengths the active inference right now is trying to move towards the the social dynamics it's doing a great job but there's still some way to go and in order to do so it doesn't have to reinvent the wheel on the other hand social sciences have done a lot of theorizing but it's it's difficult to assume causality it's difficult to see them the true mechanism of things without either doing a lot of qualitative work which itself has some of the the author's bias or specifically his own expertise into it which takes a lot of time and it's the the kinds of epistemic value you can draw from a lot of these studies is slightly lower and considered lower by all of the formal fields so if we can bring together all of these fields offer a wonderful tool to the social sciences we can bring it all together and bridge gaps in the in the communication of interdisciplinarity awesome any comments on here but if anyone's raising their hand good and that reminds me a lot of the sort of stem the science tech engineering math centrism or bias it's tied up with reductionism and a lot of other ideas but then we see initiatives like steam like throwing in the a for art or adding crafts or maker diy all of these different things that are at the edges just like you brought up at the fringe of the stem narrative or at the fringe of the accepted mainstream narrative and it's people who are making a value proposition for some of these areas that haven't been included in the stem conversation so to speak and there's various ways that this can represent itself complexity science or interdisciplinary work project-based learning interdisciplinary teams this is really also why it's so important to have different people's perspective on basically everything because this is the shape that we enclose as a group is everyone's perspective and that includes our training all these aspects so it's not just like a unilateral uh learning relationship where the humanities have so much to learn from machine learning scientists yes they do and it's also bi-directional so what does the bi-directional nexus look like and how are we going to make that nexus really a clean and accessible location instead of just making it a a slide like from one part of a playground to another all right also in the abstract we uh read that in 10.0 but mao again maybe what would you summarize this or i'm also just curious like

when you present it to different contexts or audiences so when you're in front of audience a versus b let's say what are two different ways that you talk about what you're doing um i i mean i tend to present it pretty straightforwardly to anyone because it's it's very simple it's essentially just saying that um we behave in ways that are predictable because our cognition about the world and our position in the world is also predictable through a bayesian framework most of our cognition is not perfect but been that definitely basin and so the beauty of active inference is that it gives us a way to model the different ways in which our cognition can be slightly biased or let's say not classically logical but definitely following a gradient of optimization towards x and this x is like minimization of free energy so through the scripts that are both um let's say simple and more formal so scripts that are weaker so semantic fields and stronger so behavioral links between some of the semantic elements we can essentially predict the behavior and the thought process of anyone as long as we have information about their position in the world cool and if i could just build a little bit on that it's kind of like the scripts are archetypes or patterns that play out on landscapes and that's kind of like water on a landscape and the landscape in this case is like a landscape of affordances a field of affordances that's things that we've talked about but it's also potentially quantitatively a landscape of free energy minimization or there are entities that we can model as if they're doing optimization in a variational bayesian framework on some landscape that we're also inferring about as investigators and then just like water on the landscape there might be a marsh where there's a lot of ways to go and it's not quite clear it's very flat there might be other parts that are like the grand canyon where if you're in the canyon you're totally in the canyon and if you're not you're totally not and so that would be like a very strong script in the grand canyon and then a marsh or like a flood plain would be like a weak script so that's one dimension and then the other dimension which we're going to come back to is this internalist versus externalist so we'll get to that in a second but that's another dimension about how active inference can integrate in 10.0 we also talked about the roadmap and it has again the structure that we know and love where you introduce another field here script theory in section two and then activate active inference in section three and then turn in the second part of the paper to an integrative discussion so if you have any comments now about the structuring of this paper feel free um so this is a structuring this very natural branstad that's this thing um and i think i think it works really well because we're still at a point where a lot of people don't necessarily understand what we mean when we talk about active it's it it doesn't feel that esoteric but when you read some of the papers about it some of the language is a bit complicated there's a lot of

equations so people get like okay no this is not for me whereas the the the main idea is really simple there's an attractor and phenotypes or people or entities tend to revolve as around the attractor in their states the attractor moves a little bit so that the states are not constantly at equilibrium that's really that's really the extent of what you have to understand to go through active cool yep there's so many uh aspects to what you just added there it's always important to provide little mini on-ramp in the paper itself to active inference without perhaps going into all the details and also it's important for the community to be formalizing and developing the real frameworks that we can refer out to so that we can have a short version for some papers like this and then have a really rich version that people know they can always come back to let's talk about these two dimensions of script variability so here is just one slide but this is people's life slash career slash decade long literature fiasco's back and forths it's great times and we're just gonna lay all those roughly out on this two by two grid and there's some images that kind of reflect a few of those dimensions or a few of those points in the phase space here but now just you know so much more about scripts than us um where are some examples of some of scripts that people might recognize or where has this been an especially contentious debate or is there a difference between field a and field b like uh disagreement that stems from one having an externalist perspective and one having an internalist perspective well it's it's not some emotion and disagreement it's just that depending on what your paradigm is you'll tend to focus on externalism and another paradigm you're tend to focus on internalism like for instance behaviorism could never have gone to an internalist perspective because it literally tries to evacuate anything that happens inside cognition it's like what's the input and the output so this would be fundamentally externalist um social studies will tend to to move through externalist perspectives because they don't necessarily also have access to what the perception from the individual is of a certain of a certain fact all you have is movements of demographics and and how these movies um manifest in the world so of course it's never as simple as that there's always internal elements and external misconceptions and external misconception elements and internal misconceptions but it's it's where you tend to focus in our paper we're showing that active inference allows us to bring both together because a generative model is a model of you and the world and the beauty of this is through the work of axel constant and uh samurai vessel we can see that the world also has a model of view so we have this dual connection that's possible through internalism and externalism a good example of an external script is institutions uh you you you have to go through certain um passages in order to get something you have to have a passport to go to another country and in an internalist script is

well how do i want to dress up for something tonight do i know what to wear am i going to wear something like it's it's probable that i'll get in the club if i wear xyz and there's some clubs that would stop me from going into the club if i don't wear the proper clothes this is both an internal script that i have as an expectation of what the club will allow me to do given xyz the weak script is we have some clusters of categories so um i know that some colors go with some things i know that some words are appropriate in some places and some words are not appropriate in other places but i don't necessarily know the order of the words or i don't necessarily know the order of the colors like i can wear pink up here and yellow down here or yellow and pink there's no necessary order or structure to what i have to do so long as i stay within these clusters a strong script is when you go to a wedding there's a series of events that will happen only one person wears white et cetera et cetera so it's way stricter as to where each element of the semantic clusters happen to fall both in time and in space cool thanks for that rj uh thank you for that explanation uh because i i when i first read the paper i was thinking that it's not so much disagreement as much as it is uh the way that the priors that different fields are using when they're approaching the subject and the subjects that they're studying so like when i thought about it was like externalist week maybe is like io psychology where it's a strong function of the the organizational culture it's much but and then maybe extraordinary externalist strong um would be closer to maybe maybe in the realm of geopolitics where it's looking at like procedures after after regime changes so yep and also the way um i've thought about this like internalist and externalist is that there's very little disagreement on what is for example people can say yes there's a virus particle that consists of lipids and protein and their society and people are moving around but then somebody who writes a review paper from the perspective of a protein biologist they're going to focus on the proteins and so it shouldn't be too surprising that because of their regime of attention and their knowledge affordances they're going to recommend some type of molecular binding agent whereas somebody who studies it from the sociological perspective knows that the proteins exist but then they're going to highlight different features and therefore going to make different recommendations it would just be off-key if somebody wrote a sociological review paper and then went into a conclusion about a protein binding molecule so that's something that we need to really understand that we're all at the same table but it's our different perspectives that can either divide or integrate us mal yes so what you're bringing is really important is the silence of causal links so it allows us to draw um causal chains right or at least what we believe is the inference in regard to the causal chains where you come from defines what is salient in order to get to a goal so that's

why scripts are not just like a point here and you get there however you want right that's that would be a very very very weak version of the script and anyone who is in the market blanket of where anyone tries to get to this point that's that's a very weak script but a strong script is well in order to get to this point there's x element that is salient um as you said earlier if someone comes from a different field and tells you no no the important causal chain is through externalist things it doesn't matter what's inside your mind that's behaviorism right they would say it doesn't matter what you think so what happens is if you don't do it you won't go through and the internalists will be well it doesn't matter if you go through what matters is how you act into the world but um you're ryan pointing out that if you're we're all at the same table we have to realize that there both of these things are kind of true what really matters is how they causally interact to get to um a final product a final point great point and here's something to add on that is within active inference we talk about causal inference uh the latent causes in the world as are hypotheses about the system and so if one person hypothesizes that there's a causal link and somebody else says i think it's not as important as you think it is or i think something else is more important then what we can say is instead of getting into an inference duel we can get into an active inference collaboration where we say great well given your hypothesis about the causal network and mine what's the optimal experiment that we can do and optimal experimentation entails an ethical structure nobody says well my optimal experiment is that we're gonna uh move a thousand people to mars and we're gonna test them with this device that's not it's not tractable and it's not within our affordances and it's not ethical so then we can say that's actually not in the distribution of experiments we're going to consider gonna download that family of experiments and so we can come to the table and ask what is going to be the most informative experiment that we can collaborate on specifically to reduce our uncertainty about our shared world our shared niche interesting stuff mao that's exactly right like we can come up with new models of governance through this by understanding that it doesn't have to be a zero-sum game it can literally be a productive game where we're just optimizing through the the world of possibles that overlap and that are meaningful for everyone involved so if you could model i'm being i'm being a little um optimistic here but if you could model everyone's scripts and we could model a shared understanding of what would be a um a desired state of the world which would be like rg we have this g then we can find the policies that optimize for um our our sets of constraints that's just that's just beautiful right there awesome shannon i have um the sort of same question i brought up earlier about in music you know what the goal is you know what this um goal state

is that you want to optimize towards in social interactions once you categorize them you know kind of what the goal state should be but if you don't know the goal state so you just have a couple different social interactions maybe you filmed them and and and you want to figure out can active inference if you if you do like an active inference model help you figure out what kind of social interaction the people in this video recording are engaging in oh okay now go for that so yeah so there's two things to this the first is because you're embedded in a culture that has a set of narratives you actually know what the what the goal is of given social interactions whether it's implicit or explicit you enter into into certain social interactions because you wish to enter in the goal or at least some polysemi of goals the second thing is if you are not sure you make inferences about what that goal might be and then you test those inferences by trying to you know optimize for a goal you might want or you um you check if your prediction was right and then you update so like it's it's this push and pull that allows you to get more information so um disambiguating reality is something is shown in active inference to be fundamentally valuable so me figuring out whether or not you're flirting with me right now can be a goal in and of itself even if i don't care like i don't want to flirt and i don't care that you're flirting but i wanna i wanna know though i wanna are you flirting is this an accurate model of the world that i have so that's one thing that active inference allows us to model in human behavior the second thing because we we have scripts that lead to certain pathways so certain policies policies lead to goals we can figure out what kind of policy you would sort of be engaged in and infer a potential goal you're leading towards if we're wrong eventually the model will update so we can see where are the attractors in certain behaviors which kind of behaviors tend to cluster towards something and based on this we could infer if we didn't already know what is the social goal of a specific action shannon yeah awesome the second part of your of of your answer um there's been a lot of like a this flurry of papers recently about using um the free energy principle in markup like it's this kind of thing in a very instrumentalist way right where where it doesn't mean that that group that we're looking at is necessarily embodying a markov blanket but we can predict things about that group if we model it that way and so if instrumental is reading for using active inference as a tool i'm wondering if it's going to keep moving in a similar way that machine learning has where this is a tool everybody has a vague concept of what machine learning is what it does um and then every company in the world has their machine learning algorithms to predict all of their customers behavior or they want to sell you some machine learning algorithm about categorizing some people you're interacting with or some data you have um is there maybe i

don't have a specific question maybe i just wonder what do you see would what how do you see active inference just in this like i guess the social script of how we use tools and how industries um adopt tools and then how just the general public perceives tools how do you see active inference compared to its percep the perception of machine learning wow great question get it now sorry sorry i'm so glad you asked the question i didn't wait like i'm gonna do some shameless plugging right here i'm organizing a colloquium on this exact question um the equilibrium is going to be about active inference and collective intelligence but the real goal is to discuss the ethical future of active inference in terms of social justice so a can we not make active inference into this tool that will mostly serve capitalistic aims and that will serve to further oppress communities that have zero hermeneutic power now so that's it oh aim number one aim number two is to show that active is this wonderful tool because it's not opaque it's transparent so we don't have a black box where it's like all of these layers of nose you're not you don't know what they do but it works so so with active inference we can see the parameters of the generated model we can see how it evolves even if we're using machine learning to change the parameters we can still see what those para parameters are so i do see active inference becoming the future of predictive technology however i also see that because it's transparent and because we are doing this work of trying to bring together fields that are thinking about these ethical questions such as what is the perspective of a an oppressed group and how can it inform our understanding of social dynamics it it models your anchored position into the this is a fundamental shift because that that's intersectionality right there if we can understand how intersectionality evolves in the world how how we can model a specific person's reality understanding it through mechanistic terms it doesn't have to become this sort of statistical noise that we have to move away and push people into categories you can be your own phenomenology we do not have to erase you and your reality impacts the world in certain ways so if we wanted to say to highlight a type of reality a lot of work is being done with autistic children or with gender non-binary like there's a lot of of people out there who whose realities are starting to emerge because they were erased before through active inference we could see like there's this island there that isn't quite responding like the rest can we can we zoom in on that can we see what's happening can we maybe stop thinking of society as well you either fit into the society's mold or you're you're moved away and can we think of society as something that has to adapt to its members fundamentally thanks for that amazing thank you for that thanks for this super interesting discussion i'll just add a few comments so one thing is that the instrumentalist reading which means using the framework like a toolkit rather

than a dictator about what the world quote is instrumentalism and tool building is what allows for a plurality of narratives to co-exist including social justice and non-social justice versions so because there's a plurality of narratives that means there's going to be people who are using the tool who you disagree with just like linear models and just like trains so it's going to be something that includes a big tent um rj i'll get to you in a second the second thing is that the goal of society is really as you pointed out it's polysemis there's a lot of features to it and it just made me think what is the optimal skydiving height and what's the optimal skydiving policy and for some people because of cognitive diversity it's going to be like no just not for me but no regulation or not for me not for anyone or there shouldn't be any regulation and taking me to the top or i just like two feet tall every quadrant is going to exist so that is the richness of our culture and i am very hesitant as many people should be about attempts to quantify goals for and the last point is that what active inference tools could allow would be better definition of institutions public private or new combinations so for example one organization could say our affordances are this this is our input this is our output and this is going to be a currency arbitrage mechanism that's it we're not going to be able to censor this other domain of your life on this other issue and so modularity and flexibility in will become increasingly important and again that will allow for new narratives to diversify and that will be challenging as well as uh embracing for some rj and then mao uh i was gonna say two things um so so one um on uh how much transparency it offers to to give us um i think that from from what i've seen and i'm i'm not an expert in these areas but from what i've seen on on its on the approach toward applying active in machine learning models um has been that it offers a lot more auditability that's that's one thing because the causal change and then second uh is what's basically echoing what dan just said which is that i don't know that anybody will have any control over um other people's use of the tool so i'm sure it's already being applied in a lot of these models so i'm sure that it will end up being applied to do things that most of us wouldn't like but i think that it's commendable that you're taking efforts to give people on-ramps toward using it in positive directions awesome mao that's exactly it like while you can't stop a multi-billion dollar company from using the system um you can make it so that people who usually do not have the means to partake in the construction of a tool to become salient voices in the field one of the things that tends to annoy me when i go to a conference is mammals like it seems silly right but a mantle is just mostly one type of and it doesn't to people who are on the panel it doesn't seem that way to them they're all different people with different perspectives but the way that you came into the world the way that you approach the research the way that you

approach paradigms will have some attractive attracting commonalities because of your position and a lot of brilliant work has been done in shifting paradigms because new perspectives were coming into the field intersectionality probably could only have been possible because of kimberly crenshaw who came in in feminism from a very different point the second idea is that we don't have to necessarily control what the goal state is at every level but we can probably say that um there has to be bottom-up and top-down communication for things to flow more smoothly right now highly hierarchical system tend to go you know top-down and less bottom-up the communication is a bit bottlenecked in one direction and not the other through active inference we could allow governance choose to um allow a better modeling of what is the optimal goal of ex actor and how can we model our vision of this local world to take into consideration all of these optimal goal states is it possible for us to optimize all or a maximum of their goal states um given where they are and what we have at our disposition cool one note there on the bottom up in the top down and about including people in the conversation is that uh you're really talking about a reimagining of this nexus of people and institutions it's a multi-scale re-understanding about who and what we are and we're not going to go into it in this stream but active inference and free energy has been applied to sub organismal things like the brain and different organs and physiology so we're not going to really talk about that but cells are also doing this kind of thing so it also allows us to look inwards and more internally and outwards and more externally without falling into some of these traps and then as far as um how we can make that nexus a better place the institutions and the power and the money can be pressured to stay on the rails and can be uh asked to have more transparent modeling or say if you're going to use machine learning on my preferences then i better know x y or z or if you're going to regulate water you better make promises x y and z and then also as far as rising up from the bottom and making the bottom up side of the communication you mentioned governance there's also a lot of cool work with digital governance related to active inference and then also just simply including people in relationships and conversations and teams because that is what the social fabric is so to not do that is to not be doing it and to do it is to be doing it so that's what i hope would be a cool way for people to be able to come together and talk about through these frameworks and linear models though they explain some variants in some situations you don't get this kind of conversation with a linear model so even if it does fall to pure instrumentalism with active inference and it's just a way to degenerative models that explain more variation in the environment and future models might explain even more maybe like a super neural network

that's super next level with a physical computer it might explain more variation but if it can't provide this kind of a transdisciplinary discussion then it won't be as cool as active blue hi so just to kind of piggyback on like what everybody's like been saying so i wonder like active inference is more transparent right than say like deep like neural networks but how is that like transparency affected when we don't really know like at every level it's like there's this nested hierarchy and so we don't know the affordances of all the components at every level going upward or downward in the chain of command and so i just wonder like how we can use active inference to maybe like when we don't know like maybe that's where we find the edges of the um scripts right like so like if we're looking at bottom up or top down is that like how we probe those edges for transparency when we don't know like or without transparency when we don't know what's happening at all the different levels i don't know if that makes sense if that was just confusing yeah um just rj one and then i'll first though the transparent idea that came from the active um previous one where we talked about transparent concepts in philosophy like the ones that are the water that we swim in so now it's a different concept with the auditability of machine learning but similarly there's this idea about making the invisible visible and one suggestion i'd have before rj them out would be we aren't just coming to terms on the remember also that active means that we're coming to terms on optimal experimentation so we might say all right whatever inference you're making about the world the danger of some group in another country what is the policy and what is the experimentation what's the info gain we're looking for we don't need to go for the ultimate teleos goal of society i want this to be the case you can just say okay given the info on the ground in this other country what is the policy that you and i are going to continue and deepen our relationship over and that's what's way more important is that moving forward together on a journey and on a narrative of togetherness rather than coming to agreement on these goals and i get it people want to talk about goals but they're also things that can prevent us from thinking about policy too deeply but justifying it so clearly important so yeah rj and then uh just responding to what blue said uh and it's kind of making me rethink what i said earlier about auditability it sounds like uh from what i've read that it's less getting rid of a black box and more creating a blanket of black boxes so we know where the edges between these black boxes are but that allows us to get a better understanding of what's going on in each of them i suppose so okay wow so yeah so it all depends about in on the granularity that you're using so if you let's say that i started from each individual every person i don't model uh their i don't model their cells or their brain i model them and their behavior say and then you you scale up

and up and up and up so the level of granularity of their affordances is going to be obvious at the individual level and the higher you go the more information lost there is which also makes it more computable right like if i if i if i start modeling the the behavior of a mark of blanket and then i look at all the other marker blankets at this level i don't have to recompute every individual i have the markup like it so it depends on you and your research if you want to start by modeling every cell in everybody and then try to find the emergent complexity that's great that's good and you can have more affordances based on this i think that's why scripts are so interesting because they allow us to um model the likely affordances of someone given where they are and given a set of parameters so even if it's not their absolute affordances we can assume that there's the probability of them going towards x and now if you start modeling all of the possible scripts and how it connects to semantic grounding so what does it mean for something to be a library what what is what is the clustering that i can do then you can start seeing that some clusterings will start overlapping in ways that make it more likely for the person to you know maybe go in this direction so for instance um i want to go to the library but i live far from the library and i happen to have a car well could i fly there it's unlikely right could i walk there i mean i could but i in the scripts of today's world you tend to want to do things within one or two hours if it takes more than that it's like you're wasting your time i'm probably going to take a car if i don't have a car i'm probably going to like either call an uber because i happen to be in this shoe in in this um social demographic or i'll take a bus if i'm in a different social demographic you can also predict this based on whether you think i care about the environment so the script of someone caring for the environment probably won't take a car right they'll take like the bus or something so like you can predict the overlaps based on uh the kind of scripts that are um manifest and and measurable awesome um just a couple comments there as far as transparency often people want to know more about the details of a model and i would just offer do we really understand a quadratic model or addition or linear regression credit cards bitcoin we can use things without quote understanding them that's a story we tell ourselves about the model so i would offer that transparency could be approached from the angle of like dashboards and really studying the residuals of your model so if you're doing an online learning system and then you find oh we had this person in this canyon and the feedback surveys that we're asking them are suggesting they're not getting what they need so just you know philosophical debates aside how are we going to help this person at this time and perhaps it isn't that large of an intervention that needs to be made in that case so we can separate the perfect systems depictions goal from the helping real people like in the 100 000

people in this town there's 500 people who want to go to the library and will really benefit a lot how are we going to get these 500 and then how are we going to make it open-ended with a survey so that if people aren't part of our predicted 500 they're still able to contribute these are things that are at the intersection of the model with just some pragmatics and logistics so i would say the last mile is going to be just the details of the system and this is like the skeleton wow oh i thought you had a second point i was waiting for your second point no that was it okay well no but that let's see again this is really beautiful what you said because we can promote the resilience of a community by looking at what is the true goal of people so like you said an open-ended survey for these people there's 500 who want to go to the library and the rest don't really so but do they all want to go at x times do they all live in the same area xyz knowing this information and knowing what people actually prefer what would be their goal state allows us to enable them through our resource from the top to pull down resources that would actually serve people's purposes i think that's beautiful social aim that we could promote through a new system of allocating resources depending on people's goals and benefits yep cool very fun and exciting and active stream so i'll just go to a few of these questions many of which we will have already brought up in some guys but here we can just put them all on one page and just think about what are sort of the questions that people might have again either from the acting community looking out or from the social sciences or from the neurosciences and looking in so how does active inference unify the disparate scripts concept so people can raise their hand but all emphasize the nestedness and the hierarchical like the vertical nesting and then the heteroarticle the collective behavior the horizontal the lateral these two dimensions and the ways that we can use the framework instrumentally to distinguish different models that's a way that we can get the strong versus weak scripts variation axis and the internalist externalists kind of together by just emphasizing the coherence or non-coherence strongness versus weakness of different empirical systems and take a look at them in this multi-scale way that obviates the internal external any if people have thoughts just raise their hand otherwise i just go through the oh yeah now go for it that was a lot of questions uh the first thing that come that comes to mind is you you mentioned a hierarchical hierarchical that's kind of beautiful so think about it in terms um behaviors that lead to goals right so imagine that you have these hierarchies of goals um that both are related to time and to generality so for instance um i want to be a good person which is like the highest the largest possible script ever because like that's that's one thing that's like it's kind of binary it's like okay good bad person whatever then how do i be a good person well i can

either be a good person in relation to um norm expectation or to values that i myself crafted or values that were given to me or like there's like it breaks down and then within these values set well it breaks down again into uh societal behavioral scripts etc so again see you can see that as the higher you go the less your is a strong script it's a series of strong script that becomes this sort of cluster now you take your goal at any layer they pick any of them your goal is where you want to get because it happens to be a gate to get to like the next stage of your of your script so to get to this goal there's several possibilities at the lower levels that lead in this direction we can thus unify the different uh the different types of scripts and how we understanding through this um sort of pyramid that goes from elements to goal and we can cluster which script doesn't lead to x goal et cetera et cetera and then we can understand how um the the individual models this system has a clear understanding of the system based on what they were exposed to so their environment is what they were exposed to their con their their cognition is how they brought together this sex position internalized it and started behaving behaving into the world um i feel like i answered the question but it was a very large question so let me nice that was very cool and just one other thought there is that the goal seeking uh let's not think that that's going to put our values in the background because goal seeking is even more about values so this is actually not sidestepping the question of coming to or stating our values or probable preferences for policy this is actually censoring it and then acting within it so that's the inference part all right the next question is in which settings would an active inference script model be valuable and kind of related to that would be in that setting we can already ask what would the metadata for the script look like and also i'm just curious like are there any empirical studies that are being done in the scripts framework or what next steps are you taking in this line of research or what would excite you out to see like done in a hypothetical dream experiment so um the first thing that i think of in terms of valuable is something that relates to let me check his name rj mentioned we can use the the the scripts idea in active inference and the way we modeled it through weak and strong to understand how social behaviors tend to radicalize or how some policy brought by the top will be received by the bottom so anything related to politics or to social movements would find such a model valuable the the i'm not sure what you mean by metadata so could you be a little more precise about that yeah rj why don't you go for that and maybe give some thoughts what were you meaning there sure um so first on um using actin this this this style model um in political atmospheres talk about organizations using active inference in ways we might not like um is is governments watching uh extremist ideology um so i was i was actually gonna say one place

where i i wouldn't um i know it will be of interest to use it but i would love to see it used this way is to see how people radicalize um specifically the scripts we use to react to uh extremist content um and because i i think that from my understanding um that this type of model would be very useful in identifying when a person has become uh what we'll call it like extremist audiology friendly where they're no longer going to call it out and say you know i don't agree with that or i'm not comfortable with somebody talking about saying like domestic terrorism um so so i could see it being extraordinarily useful for for governments in that context um when it comes to metadata for scripts like this is included um so the metadata might be um you know so well the data might describe the script itself um the metadata might describe some some key attributes that aren't directly related as dan likes to say the metadata is the data and this is one of these areas where like that's definitely there's no the blur is is definitely large um so for me when i think about um how i would try to start describing the metadata for scripts it would be um does this res does this script require two people to know that it's happening so like for example um uh flirting it might be you know um one person can enact their scripts without the other person uh enacting it whereas like a script for swearing in the president of the united that is a script that that's a strong script with explicit rules that everybody has to be aware of otherwise it doesn't work um so i i think that there's like categorization values that might be associated with it and then uh yeah but anyway i think there's a lot you could go into in that so i won't go too far down the rabbit hole cool let's do blue then shannon then mao so just to piggyback off of what rj was saying about radicalization it's interesting like when did radicalization become a script right and like this is what we're talking about finding the edges and so like where does the script even exist is it like an external construct like a societal construct this is the battle between internalism and externalism but when is there like a critical mass internal scripts that develop that then it becomes this external script like you can be a radical like the first person to become radicalized like was just radical right like they just took their own initiative to to you know it was their own script it was like an internal construct and so where's the like the line really like when a script becomes an acceptable or even an alternative way like there's some kind of critical mass that has to accrue before before that happens and it's just something interesting to think about shannon then mo and then rj again i have a bit of a topic change so maybe you want to talk about that first and then ask me a question okay if it's on the same um idea mao then rj yeah so exactly so how do we figure out when someone is going to radicalize or specifically what leads to radicalization well the truth is radicalization tends to happen within

clustering so it's not just someone who unless this person has specific like um psychological issues which make them go off on their own tangent away from the cluster it tends to be clusters form around narratives right so what is let's say a meninist group what is a meninist narrative well women are gaining too many rights and they're taking the spot of men and in some men in his group this narrative becomes women are trying to take down men to enslave them and then the narrative moves forward and it moves women are trying to destroy men and then what is the acceptable answer to this well we must destroy women and this is something you can see happening through the shift of the narrative the the people inside these clusters start bringing categories together and then pinging off each other and then i'm giving you this narrative and a lot of people oh yeah you're right this is the right narrative and then somebody else with slightly different a little bit more narrative keeps pushing on and you get cluster and then people build on each other and the person on top or at least someone who's believing so much into this narrative which also identifies it with potentially violence which we could measure based on the type of narrative that's being used we will possibly be able to identify okay well this cluster of people and specifically this type of person in there is more likely to do something really bad really soon what that means for our governments that's not up for me to decide ideally i think it would be um bringing help in this in these areas and being like well maybe maybe you need to rethink your narratives maybe we can help you soften whatever it is you're going through pushing you to to get towards these narratives thanks rj uh well now covered a lot of what i was going to say um once again perfect we're definitely synced um i would just add that it is definitely one of those areas where um uh i think we people who are who are working in the topic outside of governments um to try to create the speed bumps um so that like like that not not i don't want to say speed bumps it's hard to articulate like like the the like clearly express the limitations so that um government doesn't go and take this and then apply it badly like they've with um with machine learning and say like the judicial system uh we're trying to predict flight risk and things like that they applied it very quickly and didn't really weren't really sure what they were doing and they applied it before they really understood what they were working with and obviously did a lot of damage so thanks just one last comment on this uh radical doesn't mean divergent or committed radical means from the root so i think that radicalization as a discussion should refocus us on the seed memes the germ memes in the space and the ways that generative models modelable by script and active inference could understand what leads to the adoption the phase transition of the seed memes and then the policy decisions that someone engages in are probabilistic and can be inferred

downstream of these seed memetics and so those are the premises of the narrative so if someone thinks that they're in braveheart or if they think that they're in batman or if they think that they're a different role in a movie that's really what we're getting at is what is the genre of culture what kind of movie is it a good one short one long one what is the cast going to be how are they going to treat each other we can talk about it that way and that is what has been distorted in the public eye for too long shannon and then sasha thanks that was um a really informative discussion and that metaphor really brought it all together at the end daniel um so i don't study social interactions on that large of a scale i bring people in a lab i look at how maybe their motor activity coordinates in in like postural sway so something that you're not really aware of happening but it's happening all the time you're swaying and maybe you're mirroring someone um across from you and you guys have been having a conversation of what the metadata looks like um but what is i want what does the data look like if you were going to use this active inference model at something as large scale as these social movements that you're talking um or at smaller scales like i i can see the kind of data i would look at in very small groups would be you know patterns of motor activity maybe that i'm measuring from filming the body or motor activity in a brain um but in a large social group like that would you be looking at like gps tracking a phone like we've been doing a lot of modeling of how people interact in this um pandemic and where large clusters of people are interacting and we might have like super spreading events for covid19 um or would these narratives that you've been talking about um sound like their verbal narrative so would you be looking at um social media profiles um you know twitter facebook and how honest are those data so i know that the data i'm measuring the the postural sway data my participant i mean unless they're actually swaying back and forth um intentionally like to beat music they don't have control over all of these small muscle movements in your body that are that are controlling your posture um it's really a very embodied behavior and i know that when i'm measuring it i'm getting an honest signal but if you're looking at um even gps data or if you're looking at social media narrative data the signal could like people could be intentionally signaling for others who share their view or others who don't share their view or intentionally not signaling because they're afraid of a response or all of the various things that could affect and make the make the signal that you're measuring not be honest so so what would the data look like in your that that would make your active inference model of this large social situation be closer to the truth thanks for that awesome question we're gonna do sasha then then rj yeah i guess continuing the conversation of how um i would like to see this used um thinking about scripts

and stereotypes um i guess my hope for seeing active inference as a tool is for people to think about it think about their own role in their active inference story um and that instead of just thinking about machine learning and like um getting recommendations and ads based on your preferences that you actually um shape your own story and that you are building your narrative which then leads to policies and goals and so um taking a more active role in the kinds of things that will be predicted about us based on our actions um and to try to shape that in a way that signals um something deeper uh like our values ooh mao and on the data question two yeah so um i'm gonna try and uh if i forget a question they were important i really don't wanna so if i forget please remind me the first question was what kind of data well you're right social media would be super great because what you said about the honesty of data is that well social reality is manifest reality like if you behave a certain way because you know your group behaves a certain way well it's because you've integrated what the group wants and that's the script the way we could see whether or not this is something that your cognition is entirely geared towards or is just fulfilling because xyz is how you behave in other groups so one of the things that would be difficult to highlight is if you are let's say um part of a cult right you're part of a cult and you are of no contact with anyone else it would be difficult for us to figure out whether this type of behavior is also being contested inside we could maybe model this by looking at all of the other members and what kinds of interactions some people who tend to disagree a little bit tend to have we can't know for sure and that's kind of also that's we make inferences about the states of our agents based on what they put out into the world the observations they put out you're right that social media data would not necessarily be enough though um you want to have data about what it means to be in the world even outside of the virtual world so if all you care about is online then well i mean it doesn't really matter how you move but if what you care about is scripts as a whole in society well we would use um narrative data so what are the movies how do people in the movies behave and talk what is literature what are the words that are being used in what way with what valence is the positive or negative then you would use um your rights data about how people behave together to do experiments by connecting people and doing hyper scanning data like there's all of the data that would allow us to understand how a specific semantic tends to be grouped and then tends to be connected causally all of this would allow us to derive scripts it all depends about on where you want to apply this knowledge what was the second i think that covered i think i just sort of reworded my original question but that was great i think sasha had a question and i forgot sasha yes sasha yeah not so much a question just a statement for how can

people um think about applying scripts and active inference to their own behavior and behave in ways that signal at our true goals in ways that are script and socially acceptable okay okay so um this is the really cool part of having a model that can take both higher level and lower levels and you're right that giving people power over their data would also help us figure out what their true goal is right so if you if you're like i want to take an active part in shaping my reality well essentially you're giving us what your true goal is so that's good for everyone so you're um it would be great for society if people had this control i think that's where we're trying to get towards in ethical circles where we're we're not just going to take stuff from you and then model and then you won't really have any benefit from it or if you have a benefit we define what that benefit is it's more we can show you the data that is yours we can show you the data that is aggregated from you as well and you get to decide what kind of person you want to be based on this so we show you the affordance scape around you based on your profile well now you can pick what you want to do with this affordance escape cool and just so interesting about the and then moving towards action and not just targeting the ads or just targeting the consumption but actually targeting the participation as well rj and then we'll speed through the figures oh sure uh i would just add that this is definitely an area where it feels like the data and the metadata of the descriptive data about the data uh definitely blur and also uh shannon you were talking about people not having control over how they sway et cetera and i just wanted to thank you for making me hyper aware of my own posture nice um cool well what a fun discussion these are the figures and i think for next week people can start to think about what questions they want to address and then um for now i would just denote we should talk about the figures and about how they relate to some of these more qualitative things we discussed today like maybe we could even work through one of these diagrams for flirting or for ordering a cup of coffee um figure two was a layout like a graphical layout of the internal and the external realizer and so we could ask where's the markov blanket here and what does that say about how the world is what does that say about how we could model the world what are the points where we could intervene what are the experiments that could be informative whose perspective could we include when we're thinking about this like all these big questions that are what should be at the front of our mind when we approach a system like this so that was pretty cool figure let's also come back to that and uh that's basically it in our last couple minutes we'll hear a final note so anyone raise your hand if you want to make a short note other than that though we will give a form for a survey and especially for 10.2 questions from everyone who's participating if you're watching it live or in replay add a

comment with something we could address in 10.2 and uh stay in communication and other than that let's hear some last thoughts or what was your favorite part of the discussion from anyone who wants to talk um rj then mao um i after after this doug i'm definitely all the more interested in what kind of threshold events there are for the creation of scripts and for the breaking of them awesome okay um i'm super excited to see that people are interested because when i started thinking about this and and started working on it with maxwell my thought was i'm going to be this fringe researcher that no one ever quotes because no one cares and it has the beautiful side that no one will ever criticize what i do but but i'm really excited to see that people are being um inspired by this work people start asking questions and they bring in new possibilities um people from the field of music are interested which means that there's gonna be uh way more input from a variety of fields which is going to make this super rich and i'm really thankful for your um i'm not your administration um you you led this discussion really really well and that was very pleasant to be part of thank you action stream has the best norms and the best participants i have to say no i'm just kidding but really thanks everyone for making the discussion so excellent because that's really what it is and uh we hope to see you again in 10.2 hear anyone else's thoughts see you in future active streams so thanks and we will talk to you next time